Johnny Abi Zeid Daou

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EDUCATION

University of Massachusetts Boston ***Expected May 2027***
Bachelor of Science in Electrical Engineering | **GPA:** 3.784 / 4.00
Relevant Coursework & Competencies: RF/Microwave Circuits, Signals & Systems, Digital/Analog Circuit Design, Embedded Systems, Transmission Line Theory, Power Systems, Engineering Design, Semiconductors
Honors: Dean's List, UMB Alumni Recipient

SKILLS

Software: MATLAB, Python, C, LTspice, AWR Microwave Office, AutoCAD, KiCAD , Microsoft Office
Hardware & Instrumentation: Oscilloscopes, Signal Generators, SDR, Arduino, Raspberry Pi, ESP32, PCB Design
Languages: Arabic, English, French

PROJECTS

Secure Blockchain PCB for Autonomous Robots ***In Progress***
University of Massachusetts Boston, MA

- Designing a custom PCB integrating an ESP32 and NXP SE050 secure element to add hardware-backed identity, tamper detection, and safety monitoring to a low-cost robot, targeting under 40 g, under 0.8 W, and 4.5-28 V input
- Implementing SHA-256 hashing of robot data on the ESP32, targeting 10 Hz throughput and under 20 ms per signature while keeping over 100 KB of memory free
- Developing key-revocation logic for security aspects
- Contributing across the full system architecture, including the on-chain/off-chain data design that records hashes on a blockchain to detect file tampering within 1 second as well as a PCB health monitor using ML

Microcontroller Instrumentation Design - Neopixel Piano Teacher ***Spring 2025***

- Developed an interactive Raspberry Pi piano learning system that provides real-time visual guidance for beginner users
- Programmed the system in Python to process song playback, detect user inputs, and evaluate performance during testing

Microstrip Impedance Matching Transformer ***Spring 2025***

- Designed and simulated a three-section Chebyshev microstrip impedance matching transformer (200 Ω to 50 Ω) on Rogers RT/Duroid 5880 substrate in AWR Microwave Office
- Achieved a 2.5-5.5 GHz (3 GHz) matched bandwidth with in-band return loss of about 23 dB or better, keeping the narrowest trace (16.35 mil) above the 6 mil fabrication limit

AM Radio Receiver Design ***Spring 2024***

- Designed and assembled an AM radio receiver to detect a specific broadcast signal
- Analyzed signal gain, filter response, and demodulation accuracy while troubleshooting noise and impedance mismatches
- Bridged theoretical design with hands-on implementation by adjusting the radio based off of real-life effects

Microcontroller Instrumentation Design - Air Guitar ***Fall 2023***

- Designed and built a wireless Arduino-based air guitar that maps physical gestures to audio outputs in real time
- Programmed in C to handle real-time sensor input processing and wireless signal transmission
- Implemented signal conditioning techniques to improve gesture detection accuracy and system responsiveness
- Coordinated a small team across hardware and software tasks, ensuring integration of embedded code and circuit components

ACADEMIC RESEARCH

Distributed Spectrum Sensing ***In Progress***
University of Massachusetts Boston, MA

- Developing a distributed RF sensing framework for real-time spectrum monitoring, channel occupancy estimation, and dynamic spectrum access in multi-node wireless networks
- Developing Python and Bash automation tools to configure SDR-based receiver nodes, manage network communications, and streamline experimental data acquisition and processing
- Conducting multi-node RF measurements and analyzing signal power, channel utilization, and sensing performance to evaluate distributed decision-making under varying wireless channel conditions